Chapter 2
Determinants

2.1 The Determinant of A Matrix

Objectives

- Determinant
- Minor and Cofactor
- Cofactor expansion

Determinants

- Recall that if A is nonsingular, then
  - A^{-1} exists,
  - Ax = 0 has only the **trivial (zero) solution** x = 0,
  - A is row equivalent to the identity I.
- The determinant of a **square matrix $\boldsymbol{A}$** is a **scalar, denoted as** $|A|$ (commonly computed using so-called expansion by minors).
  - $|A|$ is nonzero iff A has an inverse.
  - $|A| = 0$ iff A is singular

Logic of this section

- How to define determinant
  - Defined recursively from 1 by 1 and 2 by 2 matrix and the **minors** and **cofactors**
- How to compute determinant
  - What is minor
  - What is cofactor
  - Cofactor expansion

1×1 matrices

- A = $[a]$, then A will have an inverse $A^{-1} = [a^{-1}] = [1/a]$ if and only if $a \neq 0$.
- Thus, if we define $\det(A) = a$, then A will be nonsingular if and only if $\det(A) \neq 0$.

2×2 matrices

- Let $A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$, A is nonsingular iff A is row equivalent to I.
- Suppose $a_{11} \neq 0$, then we can test whether A is row equivalent to I by elementary row operations:

$$A \approx \begin{bmatrix} a_{11} & a_{12} \\ a_{11}a_{21} & a_{11}a_{22} \end{bmatrix} \approx \begin{bmatrix} a_{11} & a_{12} \\ 0 & a_{11}a_{22} - a_{12}a_{21} \end{bmatrix}$$

  - It will be row equivalent to I if and only if

$$a_{11}a_{22} - a_{12}a_{21} \neq 0 \quad (1)$$

- How about if a_{11}=0?

$$A \xrightarrow{R1 \leftrightarrow R2} \begin{bmatrix} a_{21} & a_{22} \\ a_{11} & a_{12} \end{bmatrix} = \begin{bmatrix} a_{21} & a_{22} \\ 0 & a_{12} \end{bmatrix}$$

  - It will be row equivalent to I if and only if $a_{12}a_{21} \neq 0$ (Equation1 when a_{11}=0)

- Thus, we define $\det(A) = a_{11}a_{22} - a_{12}a_{21}$, A is nonsingular iff $\det(A) \neq 0$

Notation

The **determinant** of a square matrix A is denoted by $|A|$.

The notation **det(A)** is also used for the determinant of A.

Example

If $A = \begin{bmatrix} 2 & 4 \\ -3 & 1 \end{bmatrix}$, then $\det(A) = \begin{vmatrix} 2 & 4 \\ -3 & 1 \end{vmatrix} = 2 \times 1 - (-3) \times 4 = 14$

$n \times n$ matrices

- The determinant of an $n \times n$ matrix A is defined recursively through **cofactors**

$$|A| = \sum_{j=1}^{n} a_{ij} A_{ij} = \sum_{i=1}^{n} a_{ij} A_{ij}$$

- where $A_{ij} = (-1)^{i+j} \det(M_{ij})$ is the **cofactor** of element a_{ij},
- M_{ij} is an $(n-1) \times (n-1)$ matrix, obtained by delete the ith row and jth column of A.
- $\det(M_{ij})$ is called the **minor** of a_{ij} of A.
- The cofactor expansion transform the determinant of an $n \times n$ matrix into the summation of n determinants of $(n-1) \times (n-1)$ matrices
- The cofactor expansion can be operated on any row or any column.
- We usually choose the row or column which has many zeros.

Theorem : Cofactor expansions

Definition

The **determinant of a square matrix** is the sum of the products of the elements of the first row and their cofactors.

$$\text{If } A \text{ is } 3\times 3, |A| = a_{11}A_{11} + a_{12}A_{12} + a_{13}A_{13}$$

$$\text{If } A \text{ is } 4\times 4, |A| = a_{11}A_{11} + a_{12}A_{12} + a_{13}A_{13} + a_{14}A_{14}$$

$$\vdots$$

$$\text{If } A \text{ is } n\times n, |A| = a_{11}A_{11} + a_{12}A_{12} + a_{13}A_{13} + \cdots + a_{1n}A_{1n}$$

These equations are called **cofactor expansions** of $|A|$.

- Here A_{ij} is denoted for the cofactor, usually, my habit is use C_{ij} for cofactor ofa_{ij}

Example

Determine the minors and cofactors of the elements a_{11} and a_{32} of the following matrix A.

$$A = \begin{bmatrix} 1 & 0 & 3 \\ 4 & -1 & 2 \\ 0 & -2 & 1 \end{bmatrix}$$

Solution

$$\text{Minor of } a_{11}: |M_{11}| = \begin{vmatrix} 1 & 0 & 3 \\ 4 & -1 & 2 \\ 0 & -2 & 1 \end{vmatrix} = \begin{vmatrix} -1 & 2 \\ -2 & 1 \end{vmatrix} = (-1 \times 1) - (2 \times (-2)) = 3$$

$$\text{Cofactor of } a_{11}: A_{11} = (-1)^{1+1} |M_{11}| = (-1)^2 (3) = 3$$

$$\text{Minor of } a_{32}: |M_{32}| = \begin{vmatrix} 1 & 0 & 3 \\ 4 & -1 & 2 \\ 0 & -2 & 1 \end{vmatrix} = \begin{vmatrix} 1 & 3 \\ 4 & 2 \end{vmatrix} = (1 \times 2) - (3 \times 4) = -10$$

$$\text{Cofactor of } a_{32}: A_{32} = (-1)^{3+2} |M_{32}| = (-1)^5 (-10) = 10$$

Examples

- Find the determinants of the following matrices:

$$A = \begin{bmatrix} 1 & 4 \\ 6 & 3 \end{bmatrix} \qquad B = \begin{bmatrix} 4 & -2 \\ -6 & 3 \end{bmatrix}$$

$$C = \begin{bmatrix} 1 & 5 & -2 \\ 2 & -1 & 4 \\ 3 & -4 & -3 \end{bmatrix} \qquad D = \begin{bmatrix} a & 0 & 0 \\ 5 & b & 0 \\ 6 & 7 & c \end{bmatrix}$$

Examples

Evaluate the determinant of the following matrix *A*.

$$A = \begin{bmatrix} 1 & 2 & -1 \\ 3 & 0 & 1 \\ 4 & 2 & 1 \end{bmatrix}$$

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Evaluate the determinant of the following matrix *A*.

$$A = \begin{bmatrix} 1 & 2 & -1 \\ 3 & 0 & 1 \\ 4 & 2 & 1 \end{bmatrix}$$

Solution

$$|A| = a_{11}A_{11} + a_{12}A_{12} + a_{13}A_{13}$$
$$= a_{11}(-1)^{1+1} \det M_{11} + a_{12}(-1)^{1+2} \det M_{12} + a_{13}(-1)^{1+3} \det M_{13}$$
$$= 1\begin{vmatrix} 0 & 1 \\ 2 & 1 \end{vmatrix} - 2\begin{vmatrix} 3 & 1 \\ 4 & 1 \end{vmatrix} - 1\begin{vmatrix} 3 & 0 \\ 4 & 1 \end{vmatrix}$$
$$= [(0 \times 1) - (1 \times 2)] - 2[(3 \times 1) - (1 \times 4)] - [(3 \times 2) - (0 \times 4)]$$
$$= -2 + 2 - 6$$
$$= -6$$

Example: 4x4 determinants

Find |A|, where $A = \begin{bmatrix} 1 & 0 & -1 & 1 \\ 2 & -1 & -1 & 2 \\ 0 & -3 & 1 & 0 \\ -3 & 3 & 1 & 0 \end{bmatrix}$

- Cofactor expansion along the first row: need to calculate 3 more 3x3 determinants (each will expand into 3 more 2x2 determinants…)
- The 4th column of A has two 0's. Can we calculate determinants by cofactor expansion along this column instead? Will we still get the same answer?

Theorem: Cofactor expansion

The determinant of a square matrix is the sum of the products of the elements of any row or column and their cofactors.

- i^{th} row expansion: $|A| = a_{i1}A_{i1} + a_{i2}A_{i2} + \cdots + a_{in}A_{in}$
- j^{th} column expansion: $|A| = a_{1j}A_{1j} + a_{2j}A_{2j} + \cdots + a_{nj}A_{nj}$

Example Find the determinant of the following matrix.

$$A = \begin{bmatrix} 1 & 2 & -1 \\ 3 & 0 & 1 \\ 4 & 2 & 1 \end{bmatrix}$$

$$\begin{vmatrix} 0 & 2 & 3 & 0 \\ 0 & 4 & 5 & 0 \\ 0 & 1 & 0 & 3 \\ 2 & 0 & 1 & 3 \end{vmatrix}$$

Theorem: Cofactor expansion

The determinant of a square matrix is the sum of the products of the elements of any row or column and their cofactors.

- i^{th} row expansion: $|A| = a_{i1}A_{i1} + a_{i2}A_{i2} + \cdots + a_{in}A_{in}$
- j^{th} column expansion: $|A| = a_{1j}A_{1j} + a_{2j}A_{2j} + \cdots + a_{nj}A_{nj}$

Example Find the determinant of the following matrix.

$$A = \begin{bmatrix} 1 & 2 & -1 \\ 3 & 0 & 1 \\ 4 & 2 & 1 \end{bmatrix}$$

$$\begin{vmatrix} 0 & 2 & 3 & 0 \\ 0 & 4 & 5 & 0 \\ 0 & 1 & 0 & 3 \\ 2 & 0 & 1 & 3 \end{vmatrix}$$

Solution

$$\begin{aligned} |A| &= a_{21}A_{21} + a_{22}A_{22} + a_{23}A_{23} \\ &= -3\begin{vmatrix} 2 & -1 \\ 2 & 1 \end{vmatrix} + 0\begin{vmatrix} 1 & -1 \\ 4 & 1 \end{vmatrix} - 1\begin{vmatrix} 1 & 2 \\ 4 & 2 \end{vmatrix} \\ &= -3[(2\times 1) - (-1\times 2)] + 0[(1\times 1) - (-1\times 4)] - 1[(1\times 2) - (2\times 4)] \\ &= -12 + 0 + 6 = -6 \end{aligned}$$

Example

Solve the following equation for the variable x.

$$\begin{vmatrix} x & x+1 \\ -1 & x-2 \end{vmatrix} = 7$$

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Solution Expand the determinant to get the equation

$$x(x-2)-(x+1)(-1)=7$$

Proceed to simplify this equation and solve for *x*.

$$x^2-2x+x+1=7$$
$$x^2-x-6=0$$
$$(x+2)(x-3)=0$$
$$x=-2 \text{ or } 3$$

There are two solutions to this equation, $x = -2$ or 3.

Example

Evaluate the determinant of the following 4×4 matrix.

$$\begin{bmatrix} 2 & 1 & 0 & 4 \\ 0 & -1 & 0 & 2 \\ 7 & -2 & 3 & 5 \\ 0 & 1 & 0 & -3 \end{bmatrix}$$

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Evaluate the determinant of the following 4×4 matrix.

$$\begin{bmatrix} 2 & 1 & 0 & 4 \\ 0 & -1 & 0 & 2 \\ 7 & -2 & 3 & 5 \\ 0 & 1 & 0 & -3 \end{bmatrix}$$

Solution 3rd column expansion:

$$|A| = a_{13}A_{13} + a_{23}A_{23} + a_{33}A_{33} + a_{43}A_{43}$$

$$= 0A_{13} + 0A_{23} + 3A_{33} + 0A_{43}$$

$$= 3\begin{vmatrix} 2 & 1 & 4 \\ 0 & -1 & 2 \\ 0 & 1 & -3 \end{vmatrix}$$

$$= 3 \times 2 \begin{vmatrix} -1 & 2 \\ 1 & -3 \end{vmatrix} = 6 \times (3 - 2) = 6$$

Determinants of Triangular Matrices

Theorem 1 If A is an $n \times n$ **triangular** matrix, then the determinant of A equals the product of the diagonal elements of A.

Idea: expansion along rows or columns with lots of 0's.

Example: $\begin{vmatrix} 2 & 0 & 0 \\ 10 & 3 & 0 \\ 100 & 1000 & 4 \end{vmatrix} = 2 \begin{vmatrix} 3 & 0 \\ 1000 & 4 \end{vmatrix} = 2 \times 3 \times 4$

- Directly apply the cofactor expansion
- By induction: suppose the statement holds for $(n-1) \times (n-1)$ matrices, that is, the statement is true for its **minors.**

Properties of Determinants

Theorem 2

- If A has a row or column consisting entirely zeros, then $\det(A) = 0$.
- If A has two identical rows or columns, then $\det(A) = 0$.

Expansion according to the ith row

$$|A| = \sum_{j=1}^{n} a_{ij}A_{ij} = \sum_{j=1}^{n} a_{ij}(-1)^{i+j}\det(M_{ij})$$

- Directly apply the cofactor expansion
- By induction: suppose the statement holds for $(n-1) \times (n-1)$ matrices, that is, the statement is true for its minors.